

Education

- Mar 2021 **Doctor of Science**, *Department of Mathematical and Computing Science, Tokyo Institute of Technology*
Thesis: Difference equations arising from cluster algebras
Supervisor: Yuji Terashima and Sakie Suzuki
- Mar 2018 **Master of Science**, *Department of Mathematical and Computing Science, Tokyo Institute of Technology*
Supervisor: Yuji Terashima
- Mar 2016 **Bachelor of Science**, *Department of Information Science, Tokyo Institute of Technology*

Teaching

- Semester 2, 2025–2026 Lecturer, University College Cork, Measure Theory and Integration
- Fall 2021 Part-time lecturer, Tokyo University of Agriculture and Technology, Calculus

Fellowships

- 2025–present Postdoctoral Researcher, University College Cork
- 2024–2025 Postdoctoral Researcher, University College Dublin
- 2021–2024 JSPS Research Fellow (PD), Japan Society for the Promotion of Sciences
- 2018–2021 JSPS Research Fellow (DC1), Japan Society for the Promotion of Sciences

Preprints

- Yuma Mizuno. *Rotation fixed points of undecorated braid varieties and fusion rings of affine Lie algebras*. Oct. 1, 2026. arXiv: 2610.02034 [math.RT].
- Yuma Mizuno. *The three Kanade-Russell identities modulo nine*. Sept. 9, 2026. arXiv: 2609.09653 [math.NT].
- Yuma Mizuno. *Mutation Sequences along Weaves and Amalgamation of Braid Varieties*. Sept. 5, 2026. arXiv: 2609.05833 [math.AG].
- Tsukasa Ishibashi and Yuma Mizuno. *Adjoint Reidemeister torsion of 3-manifolds with torus boundary for semisimple algebraic groups*. 2026. arXiv: 2603.00816 [math.GT].
- Matthew de Courcy-Ireland, Matthew Litman, and Yuma Mizuno. *Divisibility by p for Markoff-like Surfaces*. 2025. arXiv: 2509.02187 [math.NT].

Publications

- Yuma Mizuno. “Periodic Y -systems and Nahm sums: the rank 2 case”. In: *SIGMA Symmetry Integrability Geom. Methods Appl.* 21 (2025), p. 094. DOI: 10.3842/sigma.2025.094.
- Yuma Mizuno. “Remarks on Nahm sums for symmetrizable matrices”. In: *Ramanujan Journal* 66 (2025), p. 62. DOI: 10.1007/s11139-025-01033-6.
- Yuma Mizuno. “ q -Painlevé equations on cluster Poisson varieties via toric geometry”. In: *Selecta Mathematica. New Series* 30 (2024), p. 19. DOI: 10.1007/s00029-023-00906-2.

Yuma Mizuno. "Difference equations arising from cluster algebras". In: *Journal of Algebraic Combinatorics* 54.1 (2021), pp. 295–351. DOI: 10.1007/s10801-020-00978-9.

Yuma Mizuno. "Exponents associated with Y -systems and their relationship with q -series". In: *SIGMA Symmetry Integrability Geom. Methods Appl.* 16 (2020), p. 028. DOI: 10.3842/sigma.2020.028.

Yuma Mizuno. "Jacobian matrices of Y -seed mutations". In: *Advances in Applied Mathematics* 115 (2020), p. 101987. DOI: 10.1016/j.aam.2019.101987.

Akishi Kato, Yuma Mizuno, and Yuji Terashima. "Quiver mutation sequences and q -binomial identities". In: *International Mathematics Research Notices* 2018.23 (2018), pp. 7335–7358. DOI: 10.1093/imrn/rnx108.